

Finding the Sweet Spot: An Empirical Study on Dataset Size, Performance, and Efficiency in Relation Extraction

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Abstract—Relation extraction (RE) is a crucial task in natural language processing that identifies relationships between entities within text. Although increasing the size of annotated datasets typically enhances model performance, the cost and effort associated with human annotation can be prohibitive. This study examines the impact of dataset size on RE performance using the Semantic Evaluation 2010 Task 8 dataset. Four deep neural models—Attention-Based Convolutional Neural Network, Attention-Based Bidirectional Long Short-Term Memory Network, Relation Bidirectional Encoder Representations from Transformers, and Matching the Blanks—were evaluated across eight subsets of the dataset to simulate size reduction. Performance metrics such as accuracy, precision, recall, and F1-score were analyzed. The results indicate that while larger datasets improve model performance, the advantages diminish beyond a certain size, illustrating the law of diminishing returns. These findings underscore the importance of balancing dataset size and quality for cost-effective training and suggest strategies such as data augmentation and transfer learning to optimize performance with limited data.

Index Terms—Relation Extraction, Language Model, Deep Learning, Diminishing Returns

I. INTRODUCTION

The size of a dataset has a significant impact on the performance of machine learning models. Larger datasets generally improve classification accuracy, while smaller datasets are at a higher risk of overfitting [1], [2]. Data collection presents various challenges, including privacy concerns, limited samples for rare cases, and legal restrictions. Additionally, working with large datasets requires more time and computational resources for training, which may be constrained [3], [4]. Despite ongoing efforts, there is still no clear definition of a small dataset. Some frameworks estimate the minimum number of samples needed for accurate results, while others rely on algorithmic information theory or establish size ranges based on prior research [5]–[7].

Prior studies on small datasets have established size ranges and highlighted their challenges for machine learning. Small datasets often lack sufficient details, making it difficult for classification algorithms to detect patterns and increasing the risk of overfitting, even to the validation set [8]. As a result,

classification models trained on small datasets may become unreliable and biased. Efforts have been made to improve classification accuracy on small datasets; however, further research is needed to understand how dataset size impacts algorithm performance and to develop robust trend identification and model development methodologies.

Recent efforts to address the challenges of small datasets have focused on enhancing classification model performance and augmenting training sets. Techniques include generating virtual samples, leveraging fuzzy rules, and employing similarity-based algorithms to create or extend dataset attributes. Methods like particle swarm optimization and Gaussian-based data generation have demonstrated improved results, emphasizing smoothness and attribute relationships. Additionally, extending attribute information has been shown to improve learning performance and mitigate data sparsity in small datasets [9]–[15].

Research has primarily focused on improving the accuracy of classification algorithms on small datasets, with studies exploring how dataset size impacts performance across domains such as sentiment classification, object detection, plant disease classification, medical classification, and information retrieval [16]–[20].

This study investigates the effect of dataset size on supervised machine learning models for Relation Extraction (RE), a task that identifies relation tuples from text. Using the Semantic Evaluation (SemEval)-2010 Task 8 dataset [21], we conducted experiments on four deep neural network models, creating eight smaller subsets to simulate size reduction. Model performance metrics were analyzed, including accuracy, precision, recall, and F-score.

The contribution of this paper is evaluating the impact of dataset size on supervised machine learning models for RE, using subsets to simulate size reduction. It compares four deep neural network models—Attention-Based Convolutional Neural Networks (Att-CNN) [22], Attention-Based Bidirectional Long Short-Term Memory (Att-BiLSTM) [23], Relation Bidirectional Encoder Representations from Transformers (R-BERT) [24], and Matching the Blanks (MTB) [25]—across

various dataset sizes, using metrics such as accuracy, precision, recall, and F-score. Focusing on the SemEval-2010 Task 8 dataset, the study examines how model performance changes with dataset size and identifies challenges in optimizing models with limited data. The findings highlight diminishing returns as dataset size increases.

The structure of this paper is as follows. Section II provides an overview of the classification models used in this study. Section III details the methodology, including dataset partitioning and sub-dataset selection. Section IV presents the experimental results, analyzing the impact of dataset size on model performance and highlighting key trends such as diminishing returns as dataset size increases and the importance of dataset size thresholds. Finally, Section V concludes the study, summarizing the findings.

II. CLASSIFICATION MODELS

We applied four different deep neural classification models, including attention-based neural networks and a deep transformer model designed to deal with the RE problem.

A. Relation Classification from Attention-Based Neural Networks

This section describes two attention-based neural networks that extract semantic relations between entities. The attention mechanism is crucial in RE, as not all words equally contribute to the semantic relationship. Identifying key clues that define the relation is essential. We will focus on two models: Att-CNN and Att-BiLSTM networks.

1) *Att-CNN*: This architecture incorporates an attention mechanism within Convolutional Neural Networks (CNNs) to identify words most relevant to the target entities and merge their representations into a sentence vector. Numerical representations are generated by combining word embeddings, position embeddings, and part-of-speech tag embeddings. The sentence convolution vector module uses a convolutional layer to aggregate local features for global relation prediction. A non-linear function generates a feature map and a max-pooling operation extracts the most essential features. The attention-based context vector module employs an attention mechanism to quantify the contextual importance of words concerning target entities directly. Each word and entity in the sequence is processed through a multi-layer perceptron to compute attention weights.

The concatenation of each entity and word representation generates a new word representation, producing a sentence representation for each entity based on attention scores that highlight informative words. The sentence convolution vector and the attention-based context vectors are then merged into a fixed-length feature vector and passed to the output layer for classification.

2) *Att-BiLSTM Networks*: This architecture leverages an attention mechanism to identify words with the closest semantic relationship to the target entities, using outputs from a Bidirectional Long Short-Term Memory (BiLSTM) network without

external knowledge. Unlike standard Long Short-Term Memory networks, which process sequences in a single direction and ignore future context, BiLSTM enhances performance by incorporating past and future context through hidden-to-hidden links flowing in both time orders. This enables the model to utilize comprehensive sequence information effectively.

The model takes a sentence as input, processed in the embedding layer to generate low-dimensional vectors for each word. These vectors are passed through a BiLSTM layer that extracts high-level features by combining forward and backward sequence contexts. The outputs are merged using an element-wise sum operation. An attention layer then assigns weights to word-level features, creating a sentence-level feature vector. Finally, using the log-likelihood loss function, a softmax classifier predicts the label based on the hidden representation.

B. Relation Classification from Deep Transformer Model

The goal is to classify relation statements in a supervised manner using pre-trained language models. This study employs the R-BERT model, which enhances relation classification by integrating pre-trained Bidirectional Encoder Representations from Transformers (BERT) embeddings with target entity information, and the MTB model, which introduces the MTB objective to refine relation representations beyond standard BERT pre-training.

1) *R-BERT*: R-BERT enhances RE by incorporating entity location information into the pre-trained BERT model. Unique tokens are added for each entity pair, including this information in the sentence vector representation.

2) *MTB for Relation Learning*: MTB improves relation representation using a novel MTB objective. The model uses entity marker tokens in BERT inputs and extracts fixed-length relation representations based on entity start states. It calculates semantic similarity scores between relation statements, identifying related or unrelated pairs without requiring predefined ontologies or labeled training data. Despite limited training data, MTB demonstrated strong performance on the SemEval-2010 Task 8 dataset.

III. METHODOLOGY

In our research into the influence of dataset size on RE performance, we adopted an experimental approach. We specifically chose the SemEval-2010 Task 8 dataset and partitioned it into eight distinct groups for analysis. We extracted seven subsets randomly using sampling from the original dataset, which is documented in Table 1. Details about the partitioning protocol are described in Algorithm 1. The objective is to investigate how the classification performance is affected by increasing the size of the same dataset. The classifiers' performance is evaluated in terms of accuracy, precision, recall, and F-score. After preprocessing, four neural network models were trained on all subsets of the original dataset. The following subsections discuss the dataset and sub-dataset selection, classification models, and performance assessment metrics.

Algorithm 1 Dataset Partitioning for Incremental Evaluation

Require: SE % SemEval-2010 Task 8 Dataset**Ensure:** $SE.1, SE.2, \dots, SE.8$ % Eight Subsets of Original Dataset% Define subset sizes: $Subset_Sizes \leftarrow [1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1]$

% Set random seed for reproducibility:

Set Random Seed

for $i = 1$ to 8 **do** $X \leftarrow Subset_Sizes[i]$ % Get size of the subset $SE.i \leftarrow$ Uniformly Random Sample of Size $X \times |SE|$ from SE **end for****End**

The SemEval-2010 Task 8 dataset is widely used for supervised RE. It contains 8,000 training and 2,717 test entity pairs, sourced from web sentences and classified into nine bidirectional relation types. These types were selected to minimize semantic overlap while covering a broad range of relations, including abstract categories like Product-Producer and Content-Container. The dataset was provided as part of a shared challenge on RE, with nine relation types and an “Other” category.

Dataset Notation	Portion	Number of instances
SE.1	1/128	62
SE.2	1/64	125
SE.3	1/32	250
SE.4	1/16	500
SE.5	1/8	1000
SE.6	1/4	2000
SE.7	1/2	4000
SE.8	1	8000

TABLE I: Dividing the SemEval-2010 Task 8 Dataset to smaller subsets.

IV. RESULTS AND DISCUSSION

This section evaluates the performance of four classifiers: Att-CNN, Att-BiLSTM, R-BERT, and MTB, using subsets of the SemEval-2010 Task 8 dataset. These subsets, labeled SE.1 to SE.8, represent increasing fractions of the dataset, from 1/128 to the full size. Metrics including accuracy, precision, recall, and F1-score were analyzed to assess model effectiveness. Att-CNN demonstrated consistent improvement as the dataset size increased, with accuracy rising from 0.22 on SE.1 to 0.77 on SE.8. Similarly, precision, recall, and F1-score followed this trend, with the latter improving sharply from 0.16 on SE.1 to 0.79 on SE.8, highlighting the model’s reliance on larger datasets for optimal performance.

Att-BiLSTM showed the most substantial gains, starting with an accuracy of 0.30 and an F1-score of 0.25 on SE.1 and achieving 0.79 and 0.80, respectively, on SE.8. Notable improvements occurred between SE.3 and SE.5, where dataset size appears critical for this model’s performance.

R-BERT, initially underperforming, saw rapid improvement as dataset size increased, achieving an accuracy of 0.84 and an F1-score of 0.81 on SE.8, with significant gains between SE.4 and SE.6 due to its transformer-based architecture.

MTB followed a similar trajectory, starting with an F1-score of 0.03 on SE.1 and improving steadily to 0.80 on SE.8, relying heavily on larger datasets for effective performance.

The results reveal a consistent trend across all classifiers: substantial performance gains occur as dataset size increases from SE.1 to SE.4, representing the steepest learning curve. Beyond SE.6 (50% of the dataset), improvements plateau, demonstrating the concept of diminishing returns. This trend holds across all metrics and classifiers, regardless of complexity, from simpler models like Att-CNN and MTB to more advanced architectures like R-BERT and Att-BiLSTM. The SE.6 subset marks a practical threshold for balancing performance improvements with computational and annotation costs, emphasizing the critical relationship between data availability and model efficacy.

Despite having more than half of the dataset available beyond the green line in Figures 1-2, none of the models—Att-CNN, Att-BiLSTM, R-BERT, or MTB—show significant performance improvement. This suggests a saturation effect, where additional data no longer contributes meaningfully to model accuracy. This could reflect architectural limitations for the Att-CNN and Att-BiLSTM models, which we need to understand better for future improvements. The flattening curve indicates that the pre-trained BERT models have already captured the key patterns in the data. The plateau we observe implies that, beyond a certain point, factors other than dataset size, such as model complexity or task-specific challenges, become more critical in driving performance improvements.

V. CONCLUSION

This research provides a comprehensive evaluation of the impact of dataset size on the performance of relation extraction (RE) systems, utilizing four distinct classifiers: Att-CNN, Att-BiLSTM, R-BERT, and MTB, applied to eight subsets of the SemEval-2010 Task 8 dataset. The findings reveal a consistent pattern across all classifiers and performance metrics: increasing the dataset size initially leads to significant performance gains; however, these improvements diminish once the dataset size exceeds a certain threshold. This phenomenon of diminishing returns underscores the importance of balancing dataset size and quality to achieve cost-effective model training. Furthermore, the study highlights the universal nature of this pattern across different architectures and evaluation metrics,

TABLE II: Performance metrics (Accuracy, Precision, Recall, and F-score) of different classification models (Att-CNN, Att-BiLSTM, R-BERT, and MTB) evaluated across eight subsets of the dataset (SE.1 to SE.8).

Classification Model	Performance Metric	SE.1	SE.2	SE.3	SE.4	SE.5	SE.6	SE.7	SE.8
Att-CNN	Accuracy	0.22	0.26	0.37	0.48	0.58	0.66	0.73	0.77
	Precision	0.21	0.16	0.26	0.41	0.57	0.64	0.74	0.79
	Recall	0.20	0.25	0.30	0.41	0.59	0.69	0.74	0.79
	F-score	0.16	0.19	0.27	0.40	0.56	0.66	0.74	0.79
Att-BiLSTM	Accuracy	0.30	0.45	0.46	0.60	0.66	0.72	0.75	0.79
	Precision	0.34	0.42	0.42	0.59	0.65	0.70	0.76	0.81
	Recall	0.33	0.39	0.42	0.60	0.73	0.76	0.77	0.80
	F-score	0.25	0.37	0.38	0.57	0.65	0.71	0.75	0.80
R-BERT	Accuracy	0.20	0.26	0.41	0.65	0.73	0.79	0.82	0.84
	Precision	0.24	0.26	0.34	0.65	0.71	0.75	0.77	0.79
	Recall	0.13	0.12	0.28	0.56	0.69	0.77	0.81	0.83
	F-score	0.07	0.11	0.27	0.54	0.69	0.76	0.79	0.81
MTB	Accuracy	0.19	0.22	0.38	0.59	0.71	0.78	0.83	0.85
	Precision	0.19	0.23	0.32	0.58	0.69	0.74	0.76	0.79
	Recall	0.11	0.12	0.25	0.51	0.66	0.73	0.78	0.80
	F-score	0.03	0.09	0.20	0.51	0.67	0.74	0.78	0.80

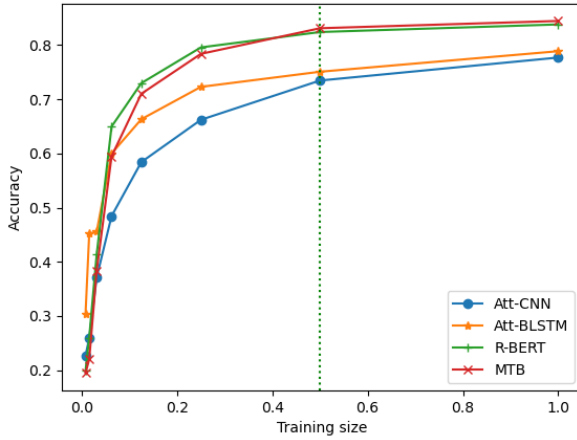


Fig. 1: Impact of training data size on the accuracy of Att-CNN, Att-BiLSTM Networks, and pre-trained BERT models (R-BERT and MTB) used for RE task.

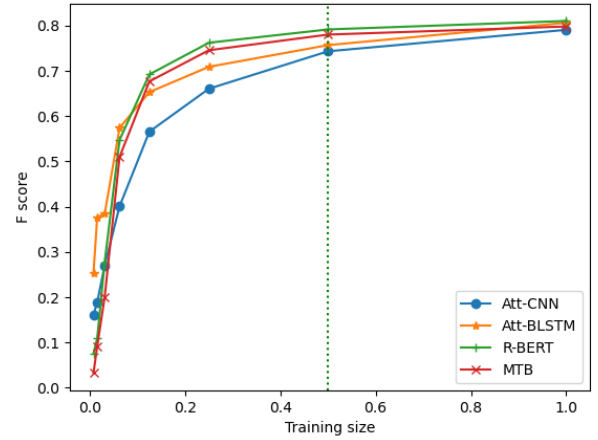


Fig. 2: Impact of training data size on the F-score of Att-CNN, Att-BiLSTM Networks, and pre-trained BERT models (R-BERT and MTB) used for RE task.

providing valuable insights for optimizing data annotation efforts in natural language processing tasks.

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